



The Papyrus Handbook

A guide to Markdown books with Papyrus

milon/papyrus

The Papyrus Handbook

Markdown to Book — PDF, EPUB, HTML, and KDP

Papyrus

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This handbook is the companion guide for **milon/papyrus** — a PHP CLI that turns a Markdown book project into PDF, EPUB, HTML, and Amazon KDP exports.

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Introduction

Papyrus is a Composer package and CLI for authors who write books in Markdown and need print-ready PDF, EPUB, single-file HTML, and Amazon KDP outputs from one project layout.

This handbook is itself a Papyrus book. The source lives in [examples/the-papyrus-handbook/](#); rendered PDF and HTML are checked into [docs/](#) so you can read without building.

What you get

Format	Typical use
PDF	Print interiors, digital reading, sample PDFs
EPUB	Stores, e-readers, KDP Kindle upload
HTML	Online reading, sharing a single file
KDP	Kindle EPUB, print PDF, cover export, metadata JSON

Project shape

Every book is a directory with:

- `papyrus.php` — title, trim, themes, Mermaid, sample ranges, KDP
- `content/` — Markdown chapters (natural filename order)
- `assets/` — PDF themes, HTML theme, EPUB CSS, covers, fonts
- `export/` — build outputs (usually gitignored)

Next steps

Install Papyrus, scaffold or copy this sample, run `papyrus doctor`, then `papyrus build`. The chapters that follow cover each step in detail.

Install and project layout

Requirements

- PHP 8.2+ with extensions `dom`, `gd`, `mbstring`, `zip`, and `zlib`
- Composer
- Optional: Node.js and `@mermaid-js/mermaid-cli` (`mmdc`) when Mermaid is enabled

Install per project

In your book repository:

```
composer require milon/papyrus
```

```
vendor/bin/papyrus --version  
vendor/bin/papyrus init  
vendor/bin/papyrus doctor
```

Wire Composer scripts:

```
{  
  "scripts": {  
    "build": "papyrus build",  
    "build:pdf": "papyrus build:pdf --theme light,dark",  
    "build:epub": "papyrus build:epub",  
    "build:html": "papyrus build:html",  
    "build:sample": "papyrus build:sample",  
    "build:kdp": "papyrus kdp",  
    "doctor": "papyrus doctor"  
  }  
}
```


Install globally

```
composer global require milon/papyrus  
export PATH="$(composer global config bin-dir --absolute):$PATH"  
papyrus list
```

Scaffold a book

```
papyrus init  
papyrus init -d my-book
```

`init` creates `papyrus.php`, `content/`, and `assets/`. Use `--force` / `-f` to overwrite.

Layout

Path	Role
<code>papyrus.php</code>	Book settings
<code>content/</code>	Markdown chapters
<code>assets/</code>	Themes, CSS, covers, fonts
<code>export/</code>	Built artifacts
<code>.papyrus/</code>	Incremental caches

Always run from the book root, or pass `-d / --dir`. Override where artifacts are written with `-e / --export` (default: `<book>/export`):

```
papyrus doctor -d /path/to/book
papyrus build --dir /path/to/book
papyrus build:html -d /path/to/book -e /path/to/out
```

Configuration

All settings live in `papyrus.php`, which returns a PHP array.

Identity

```
return [  
  'title' => 'My Book',  
  'subtitle' => 'A short subtitle',  
  'author' => 'Author Name',  
  'themes' => ['light', 'dark'],  
];
```

`themes` lists PDF theme names. Each needs `assets/theme-{name}.html`.

The export filename slug comes from the title (lowercased, hyphenated). This handbook is *The Papyrus Handbook*, so outputs look like `the-papyrus-handbook-light.pdf`.

Page size and margins

```
'document' => [  
  'size' => 'crown-quarto',  
  'margin_left' => 27,  
  'margin_right' => 27,  
  'margin_top' => 14,  
  'margin_bottom' => 14,  
],
```

List presets with **papyrus sizes**. Custom trim:

```
'document' => [  
  'format' => [152.4, 228.6], // width_mm, height_mm  
],
```

Table of contents

```
'toc' => [  
  'h1' => 0,  
  'h2' => 0,  
  'h3' => 1,  
],
```

Values follow mPDF **h2toc** conventions used by Papyrus (**0** include, **1** exclude).

Cover images

```
'cover' => [  
  'image' => 'cover.png',  
  'light' => 'cover-light.png',  
  'dark' => 'cover-dark.png',  
],
```

Paths are relative to **assets/**. Missing per-theme covers fall back to **cover.image**.

Running header

```
'header' => [  
  'style' => 'font-style: italic; text-align: right; border-  
bottom: solid 1px #808080;',  
],
```

Pretoc chapters suppress folios differently from body chapters.

Fonts

Declare faces under `assets/fonts/` and optional **script routing** (PDF only). First matching **script** rule wins.

```
'fonts' => [  
  'default' => 'librelibertine', // optional; else first face,  
  // else mPDF default  
  'faces' => [  
    [  
      'name' => 'librelibertine',  
      'regular' => 'LinLibertine_R.ttf',  
      'bold' => 'LinLibertine_RB.ttf',  
      'italic' => 'LinLibertine_RI.ttf',  
      'bold_italic' => 'LinLibertine_RBI.ttf',  
    ],  
    [  
      'name' => 'notosansbengali',  
      'regular' => 'NotoSansBengali-Regular.ttf',  
      'otl' => true, // OpenType layout – needed for  
      // complex scripts  
    ],  
  ],  
  'script' => [  
    // When mPDF tags a run as Bengali, use this face instead  
    // of the default  
    ['match' => ['bn', 'ben', 'bengali'], 'face' =>  
    'notosansbengali'],  
  ],  
],
```

Faces

Each face needs a **name** and a **regular** file that exists under **assets/fonts/**. Optional keys: **bold**, **italic**, **bold_italic**, and **otl** (sets mPDF **useOTL** when true — common for code fonts and Indic / Arabic scripts).

Papyrus merges these into mPDF's **fontdata** when building PDF (**FontRegistry** → **MpdfFactory**).

Script rules

fonts.script is a list of { **match**, **face** } rules used only for **PDF**. They do not change EPUB or HTML typography.

When any applicable rule exists, Papyrus turns on mPDF's **autoScriptToLang** and **autoLangToFont**, and installs a custom **languageToFont** resolver (**ScriptLanguageToFont**). mPDF then:

1. Detects script runs in the HTML (e.g. Bengali glyphs).
2. Tags them with a language / script code (e.g. **bn**, or ***-beng**).
3. Asks Papyrus which face to use; the first rule whose **match** list hits that code wins.

match aliases are lowercased. Built-in aliases include:

Aliases you can list	Resolves as
bn , ben , beng , bengali	Bengali

Aliases you can list	Resolves as
<code>ar, arab, arabic</code>	Arabic
<code>hi, hin, deva, devanagari, hindi</code>	Devanagari
any 4-letter ISO script code	that script as-is

The `face` string must be a name from `fonts.faces` and that face must actually load (regular file present). Rules pointing at a missing face are skipped (`FontRegistry::applicableScriptRules()`). Unmatched languages fall through to mPDF's built-in `LanguageToFont`.

Example: body text in Latin uses `librelibertine`; a paragraph containing বন্ধন is auto-tagged Bengali and rendered with `notosansbengali` if that face is registered.

Notice: Without both a registered face and a matching `script` rule, non-Latin runs often show as tofu (missing glyphs) in the default font.

Break level

```
'breakLevel' => 2,
```

Automatic page breaks before headings (**1** = H1, **2** = H1 and H2). Explicit **[break]** markers always insert a break.

CommonMark hook

```
'configure_commonmark' => [  
  // callable(s) receiving the Environment  
],
```

Writing content

Chapters are Markdown under **content/**, loaded in natural filename order (**00-...**, **01-...**, **10-...**).

Front matter

```
---
title: My chapter
pretoc: true
---

Body starts here.
```

Key	Meaning
<code>title</code>	Chapter title for EPUB / internal use
<code>pretoc</code>	<code>true</code> places the chapter before the PDF TOC

Markdown features

CommonMark + GFM, fenced code highlighting, and book extras.

Emphasis and code

```
Hello, world.  
  
`inline code`  
  
```php  
$user = User::factory()->create();
```
```

Callouts

```
> {notice} Something helpful.  
> {warning} Something risky.
```

```
::: note
A note aside.
:::

::: warning
A warning aside.
:::
```

Page breaks

```
[break]
```

Mermaid

Write a **mermaid** fence in your chapter. Papyrus runs **mmdc** at build time and replaces the fence with an SVG (or PNG) figure in PDF, EPUB, HTML, and site exports.

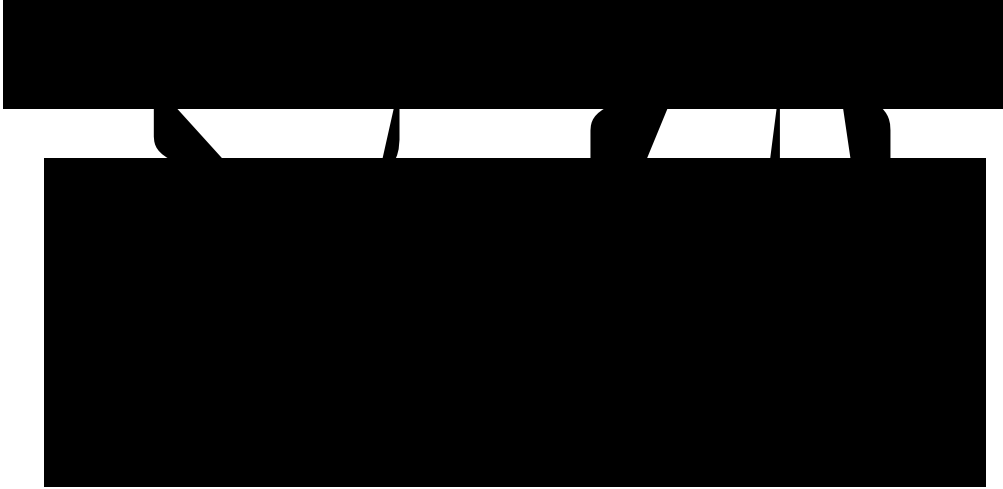
Source:

```
```mermaid
flowchart TB
 subgraph Authoring
 MD[Markdown chapters]
 CFG[papyrus.php]
 ASSETS[Themes and fonts]
 end

 subgraph Papyrus
 CONVERT[BookConverter]
 MERMAID[MermaidRenderer]
 PDF[build:pdf]
 EPUB[build:epub]
 HTML[build:html]
 SITE[build:site]
 end

 MD --> CONVERT
 CFG --> CONVERT
 ASSETS --> PDF
 ASSETS --> HTML
 ASSETS --> SITE
 CONVERT --> MERMAID
 MERMAID --> PDF
 MERMAID --> EPUB
 MERMAID --> HTML
 MERMAID --> SITE
```
```

Rendered output:



Enable in `papyrus.php` (theme defaults to `auto` — book colours; HTML and site embeds both light and dark variants):

```
'mermaid' => [  
    'enabled' => true,  
],
```

Optional knobs: `format` (`svg` / `png`), `theme` (`auto` or a Mermaid stock theme like `default` / `dark` / `forest`), `max_width_mm`.

Requires `@mermaid-js/mermaid-cli` (`mmdc`) on `PATH`. Diagrams cache under `.papyrus/cache/mermaid`.

PHP fence linting

```
papyrus lint  
papyrus lint --fix
```

Themes and assets

Everything visual lives under `assets/`. New projects from `papyrus init` ship the same PDF theme used by *The Filament Playbook* (Libre Libertine body, Times headings, OxProto code, github-gist highlight colors, notice/tip/caution asides) plus matching font files under `assets/fonts/`.

PDF themes

For each name in `themes`, provide `assets/theme-{name}.html`. Split the theme on the TOC marker:

```
<!-- PAPYRUS:TOC -->
```

Papyrus writes pretoc chapters, then the TOC, then body chapters around that marker. A legacy `<!-- IBIS:TOC -->` marker is still recognized after `migrate-ibis`.

Typical theme responsibilities: `@page` / typography, code blocks, callout colors, title page, and running header styles (often paired with `header.style` in config).

HTML theme

`assets/theme-html.html` is a full HTML document with placeholders `{{ $title }}`, `{{ $subtitle }}`, `{{ $author }}`, and `{{ $body }}`. The default template mirrors the light PDF theme (same typefaces, colors, code blocks, and callouts), with a fixed moon/sun icon toggle that switches to the dark PDF palette. The reading column is **52em** wide (a bit wider than the PDF measure). `@font-face` rules in the theme use `../assets/fonts/`; at build time Papyrus rewrites those URLs relative to the HTML output directory (so `-e docs` still finds the book's fonts).

```
papyrus build:html
papyrus build:site
```

`build:site` writes `export/<slug>-site/` with one HTML page per chapter, a Home index, shared CSS/JS, a collapsible chapter sidebar (Docsify-style), and the same light/dark toggle as the single-file HTML export.

EPUB styles

`assets/style.css` (and optional `highlight.codeblock.min.css`) ship inside the EPUB. Prefer simple `pre/code` rules for e-ink readers.

Covers and fonts

Place cover images and font files under `assets/` (fonts usually in `assets/fonts/`). Reference them from `cover` and `fonts.faces` in `papyrus.php`.

Building exports

Validate first, then build.

```
papyrus doctor  
papyrus build
```

build runs every configured PDF theme, then EPUB, HTML, and any enabled KDP outputs.

Individual formats

```
papyrus build:pdf
papyrus build:pdf --theme light
papyrus build:pdf --theme light,dark --parallel
papyrus build:epub
papyrus build:html
papyrus build:site
papyrus build:sample
```

| Command | Output (slug from title) |
|---------------------------|--|
| <code>build:pdf</code> | <code>export/<slug>-<theme>.pdf</code> |
| <code>build:epub</code> | <code>export/<slug>.epub</code> |
| <code>build:html</code> | <code>export/<slug>.html</code> |
| <code>build:site</code> | <code>export/<slug>-site/</code> (multi-page site + sidebar) |
| <code>build:sample</code> | <code>export/sample-<slug>-<theme>.pdf</code> |

Use `-e` / `--export` to write somewhere other than `export/` (for example this repo's handbook docs: `-d examples/the-papyrus-handbook -e docs`).

`--parallel` / `-p` builds each PDF theme in its own process.

Watch mode

```
papyrus watch  
papyrus watch --interval 3
```

Caching

Chapter HTML caches under `.papyrus/cache/markdown`. Mermaid figures use `.papyrus/cache/mermaid`. Delete `.papyrus/` for a cold rebuild.

Vendor noise

mPDF and the EPUB zipper sometimes emit PHP warnings. Papyrus collects third-party notices and only prints them with `-v` / `--verbose`.

Samples and KDP

Sample PDF

Carve page ranges from a full theme PDF:

```
'sample' => [  
  'ranges' => [  
    ['from' => 1, 'to' => 4],  
    ['from' => 10, 'to' => 12],  
  ],  
,  
  'sample_notice' => 'This is a sample from My Book.',
```

```
papyrus build:sample --theme light
```

Pages are 1-based inclusive ranges against the finished PDF.

Amazon KDP

```
'kdp' => [  
  'ebook' => [  
    'enabled' => true,  
    'cover' => 'cover-ebook.jpg',  
  ],  
  'print' => [  
    'enabled' => true,  
    'bleed_mm' => 3,  
    'margin_preset' => 'recommended',  
    'paper' => 'white',  
  ],  
  'metadata' => [  
    'description' => 'Bookstore blurb...',  
    'keywords' => ['laravel', 'filament'],  
    'language' => 'en',  
  ],  
],
```

```
papyrus kdp  
papyrus kdp:ebook  
papyrus kdp:print --theme light  
papyrus kdp:cover  
papyrus kdp:metadata
```

| Command | Typical artifact |
|---------------------------|--|
| <code>kdp:ebook</code> | <code>export/<slug>-kdp.epub</code> |
| <code>kdp:print</code> | <code>export/<slug>-kdp-print.pdf</code> |
| <code>kdp:cover</code> | cover assets under <code>export/</code> |
| <code>kdp:metadata</code> | <code>export/<slug>-kdp-metadata.json</code> |

Confirm trim and margins with `papyrus sizes` and Amazon's current KDP print specs before uploading.

Migration and CI

Migrating from ibis-next

```
papyrus migrate-ibis  
papyrus migrate-ibis -d /path/to/book
```

This writes `papyrus.php` and updates theme TOC markers to `<!-- PAPYRUS:TOC -->`. After migration:

1. Remove `hi-folks/ibis-next` and any Composer patches for `ibis/mPDF`.
2. Point scripts at `papyrus` (`build`, `build:pdf`, ...).
3. Run `papyrus doctor` and a full `papyrus build`.

Continuous integration

Copy the stub workflow from the Papyrus package:

```
stubs/github/workflows/book-build.yml
```

A typical book job installs PHP with `dom`, `gd`, `mbstring`, `zip`, `zlib`, runs `composer install`, `papyrus doctor`, optional `papyrus lint`, then `composer build`.

Programmatic use

```
use Milton\Papyrus\Config\Project;  
  
$project = Project::load($bookDir);  
$book =  
$project->bookConverter()->convertDirectory($project->contentDir)  
;
```

Checklist before release

1. `papyrus doctor` is clean
2. `papyrus build` succeeds for every theme
3. Sample ranges still make sense after reflows
4. KDP EPUB opens in Kindle Previewer; print PDF matches trim/bleed
5. HTML export readable on phone and desktop
6. Pin a Papyrus version in the book's `composer.json`

Command reference

| Command | Purpose |
|---------------------------|--|
| <code>init</code> | Scaffold <code>papyrus.php</code> , <code>content/</code> , <code>assets/</code> |
| <code>doctor</code> | Validate config, paths, and optional Mermaid |
| <code>build</code> | All PDF themes, EPUB, HTML, enabled KDP |
| <code>build:pdf</code> | PDF themes (<code>--theme</code> , <code>--parallel</code>) |
| <code>build:epub</code> | EPUB3 |
| <code>build:html</code> | Single-file HTML |
| <code>build:site</code> | Multi-page HTML site with chapter sidebar |
| <code>build:sample</code> | Sample PDF from <code>sample.ranges</code> |
| <code>kdp</code> | All enabled KDP outputs |
| <code>kdp:ebook</code> | Kindle EPUB |
| <code>kdp:print</code> | Print interior PDF |

| Command | Purpose |
|---------------------------|--|
| <code>kdp:cover</code> | Cover assets under <code>export/</code> |
| <code>kdp:metadata</code> | Metadata JSON sidecar |
| <code>sizes</code> | List page-size presets |
| <code>migrate-ibis</code> | <code>ibis.php</code> → <code>papyrus.php</code> |
| <code>lint</code> | Lint PHP fences (<code>--fix</code>) |
| <code>watch</code> | Rebuild on changes (<code>--interval</code>) |

Common options:

- `-d` / `--dir` — book root (default: current directory)
- `-e` / `--export` — override export directory (default: `<book>/export`)
- `-v` / `--verbose` — include third-party vendor notices

Packagist: [milon/papyrus](https://packagist.org/packages/milon/papyrus). Source: github.com/milon/papyrus.